The Cavendish Experiment

Measurement of the Gravitational Constant G with a Torsion Balance

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Abstract

The Cavendish torsion balance experiment was performed to determine an experimental value of the gravitational constant G by measuring the weak gravitational attraction between laboratory-scale masses. A torsion balance consisting of two small masses suspended by a thin fibre was used, while two larger external masses produced a gravitational torque on the system. Angular displacement of the balance was measured using an optical lever in which a laser beam reflected from a mirror on the balance was detected by a linear photodiode array. The oscillation period of the torsion system and the equilibrium angular deflection were determined from the recorded motion of the laser spot. These measurements yielded an experimental value of $G = (9.77 \pm 0.18) \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, which differs from the accepted value by approximately 46%. The discrepancy is attributed primarily to the sensitivity of the apparatus to geometric uncertainties and environmental vibrations. Despite these limitations, the experiment demonstrates the principle of the Cavendish torsion balance and the challenges associated with measuring extremely small gravitational forces.

1 Introduction

Newton's law of universal gravitation says that any two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them,

$$F = G\frac{m_1 m_2}{r^2}, \tag{1}$$

where m_1 and m_2 are the interacting masses, r is the separation between their centres, and G is the universal gravitational constant [1]. Although Newton established this relationship in his 1867 *Philosophiae Naturalis Principia Mathematica*, the value of the gravitational constant itself remained unknown for more than a century. Because gravitational interactions between laboratory scale masses are extremely weak, early scientists were unable to measure gravitational forces directly and instead relied on astronomical observations to study gravitational phenomena.

The first successful laboratory experiment capable of measuring gravitational attraction between known masses was performed by Henry Cavendish in 1798 using a torsion balance apparatus [2]. Cavendish did not frame his work as a measurement of the gravitational constant but instead as an attempt to determine the "mean density of the Earth." By measuring the extremely small gravitational attraction between known laboratory masses, Cavendish was able to infer the mass of the Earth and therefore its average density.

The relationship between the gravitational constant and the Earth's mass follows from the expression for the gravitational acceleration at the Earth's surface,

$$g = \frac{GM_E}{R_E^2}, \tag{2}$$

where M_E is the mass of the Earth and R_E is its radius. Once G is determined experimentally, the mass of the Earth can be obtained from the measured value of g. The mean density of the Earth can then be calculated from

$$\rho_E = \frac{M_E}{\frac{4}{3}\pi R_E^3}. \tag{3}$$

Cavendish's experiment therefore provided the first quantitative determination of the Earth's density and effectively established the first experimental measurement of the gravitational constant.

The Cavendish experiment uses a torsion balance consisting of a horizontal beam suspended by a thin torsion fibre with small masses attached at its ends. Two larger external masses are placed near the small masses, producing a gravitational attraction that exerts a torque on the beam [4]. The torsion fibre resists this rotation with a restoring torque proportional to the angular displacement,

$$\tau_{\text{torsion}} = k\theta, \tag{4}$$

where k is the torsion constant of the fibre and θ is the angular deflection.

The gravitational force between a large mass M and a small mass m separated by a distance r produces a torque about the fibre axis given by

$$\tau_{\text{grav}} = 2L\frac{GMm}{r^2}, \tag{5}$$

where L is the distance from the torsion fibre to the small masses. The factor of two arises because two identical mass pairs contribute symmetrically to the torque. At equilibrium the restoring torque of the fibre balances the gravitational torque,

$$k\theta = 2L\frac{GMm}{r^2}. \tag{6}$$

Solving for the gravitational constant gives

$$G = \frac{k\theta r^2}{2MmL}. \tag{7}$$

The torsion constant k of the fibre cannot be measured directly and must instead be determined from the oscillation properties of the torsion balance. For small angular oscillations the system behaves as a torsional harmonic oscillator with period

$$T = 2\pi\sqrt{\frac{I}{k}}, \tag{8}$$

where I is the moment of inertia of the rotating system. Rearranging gives

$$k = \frac{4\pi^2 I}{T^2}. \tag{9}$$

For a beam with two small masses located a distance L from the torsion axis and assuming the beam mass is negligible, the moment of inertia is

$$I = 2mL^2. \tag{10}$$

By measuring the oscillation period of the torsion system and the angular deflection produced by nearby large masses, the torsion constant of the fibre and the gravitational torque acting on the system can be determined, allowing an experimental value of the gravitational constant to be calculated.

Despite more than two centuries of experimental effort, the gravitational constant remains one of the least precisely known fundamental constants. Modern measurements using advanced torsion balances and other experimental techniques still show discrepancies larger than their reported uncertainties [3]. The Cavendish torsion balance therefore remains both historically and pedagogically significant, demonstrating how extremely weak gravitational forces between laboratory masses can be measured.

The objective of this experiment is to determine an experimental value of the gravitational constant using a Cavendish torsion balance. Measurements of the oscillation period of the torsion system and the equilibrium angular displacement produced by nearby masses are used to determine the torsion constant of the fibre and the gravitational torque acting on the beam. From these measurements the gravitational constant can be calculated and compared with the accepted value.

2 Experimental Method

2.1 Apparatus

The experiment was performed using a Cavendish torsion balance designed to measure the extremely small gravitational attraction between laboratory scale masses. A schematic of the apparatus used in this experiment is shown in Fig. 1. The torsion balance consisted of a horizontal beam suspended from a thin torsion wire. Two identical small masses were attached to the ends of the beam and enclosed within a protective housing in order to minimize disturbances from air currents and external vibrations. Two larger external masses were positioned outside the enclosure near the small masses so that the gravitational attraction between each large mass and its neighbouring small mass produced a torque on the suspended beam.

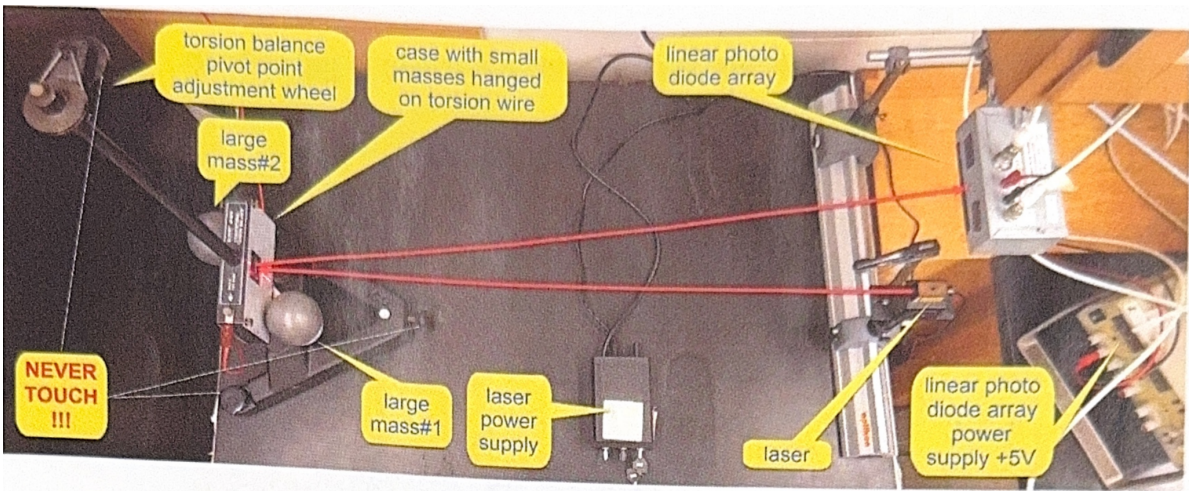


Figure 1: Cavendish torsion balance apparatus used in this experiment. The suspended beam carrying two small masses is attached to a torsion wire and enclosed within a protective housing. Two large external masses produce a gravitational torque on the system. A laser beam reflects from a mirror mounted on the torsion balance and is detected by a linear photodiode array, allowing the angular displacement of the beam to be measured [4].

When the large masses were positioned near the small masses, the gravitational interaction between each pair generated a torque about the torsion fibre. This torque caused the beam to rotate slightly until equilibrium was reached between the gravitational torque and the restoring torque of the twisted wire.

Angular motion of the torsion balance was measured using an optical lever system. A laser beam was directed toward a mirror mounted on the torsion balance assembly. The reflected beam was detected by a linear photodiode array positioned several tens of centimetres away from the apparatus. As the torsion balance rotated, the reflected laser beam moved across the photodiode array. The position of the laser spot on the detector was recorded in pixel units, allowing the angular displacement of the torsion balance to be monitored. The use of the photodiode array allowed the motion of the torsion system to be recorded continuously with high sensitivity.

2.2 Measurement Procedure

When the large masses were positioned near the small masses, the gravitational attraction between each pair of masses produced a torque on the torsion balance and shifted the equilibrium position of the suspended beam. Because the torsion wire provides a restoring torque proportional to angular displacement, the system behaves as a torsional oscillator.

If the beam is disturbed from its equilibrium position, the torsion balance undergoes small damped oscillations about this equilibrium angle. These oscillations were monitored using the optical lever system described above. As the torsion balance rotated, the reflected laser beam moved across the photodiode array, causing the recorded pixel position of the laser spot to vary periodically with time.

The oscillation period of the torsion system was determined by recording the pixel position of the reflected laser beam as a function of time and identifying successive maxima and minima of the oscillatory motion. The oscillation period was then obtained from the time interval between successive peaks in the pixel position data.

In addition to oscillation measurements, the equilibrium displacement of the torsion balance caused by the gravitational interaction between the masses was determined from the average shift in the pixel position of the reflected laser beam.

2.3 Experimental Runs

Two measurement runs were performed in order to record the oscillatory behaviour of the torsion balance under similar experimental conditions.

Prior to the first run, the apparatus was left undisturbed for approximately 45 minutes in order to allow the torsion system to reach equilibrium and to ensure that transient oscillations produced during the setup of the apparatus had sufficiently decayed. This stabilization period minimized disturbances and allowed the system to settle before measurements were recorded.

Following this waiting period, the position of the reflected laser beam on the photodiode array was recorded as a function of time. The pixel location of the laser spot was monitored continuously, producing a time series that captured the oscillatory motion of the torsion balance.

A second measurement run was then performed after repositioning the large masses. No additional stabilization period was applied prior to the second run. The same measurement procedure was repeated and the pixel position of the reflected laser beam was again recorded as a function of time. This data was later used to determine the oscillation period of the torsion system and the equilibrium deflection produced by the gravitational interaction between the masses.

3 Results

3.1 Oscillatory Motion of the Torsion Balance

The angular motion of the torsion balance was monitored through the position of a laser beam reflected from a mirror mounted on the balance arm and detected by a linear photodiode

array. The pixel position of the reflected beam therefore provided a direct measure of the angular motion of the torsion system.

Figures 2 and 3 show the recorded pixel position as a function of time for two experimental runs. In both cases the system exhibits periodic oscillations characteristic of a torsional oscillator. A gradual decrease in oscillation amplitude is observed over time, indicating the presence of damping in the torsion fibre and surrounding air.

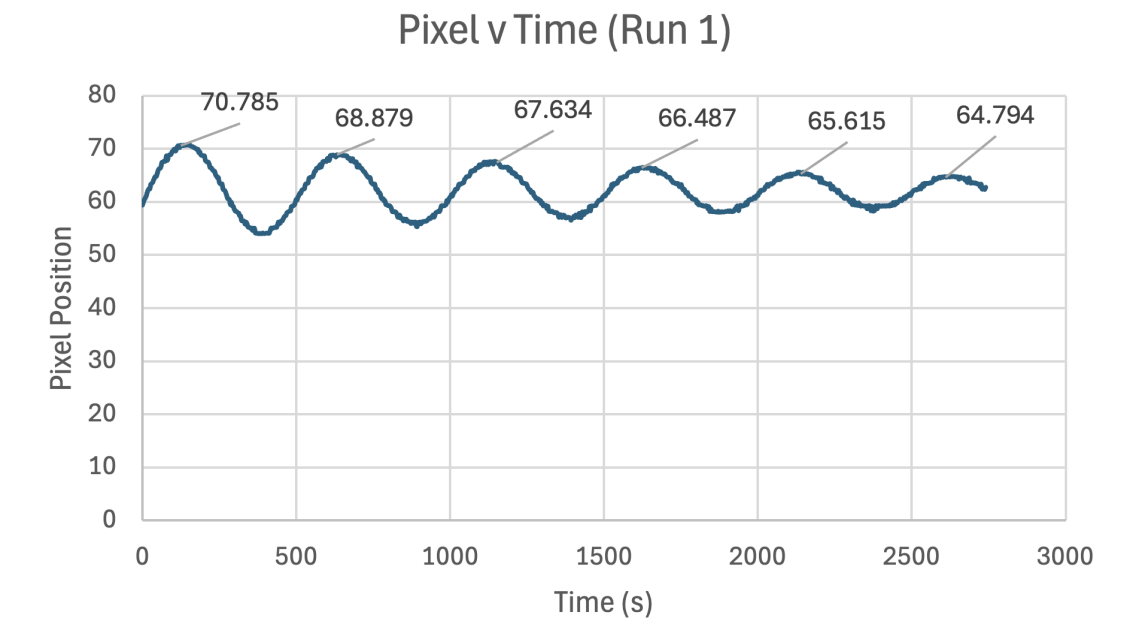


Figure 2: Pixel position of the reflected laser beam as a function of time for Run 1. The periodic variation corresponds to torsional oscillations of the balance beam.

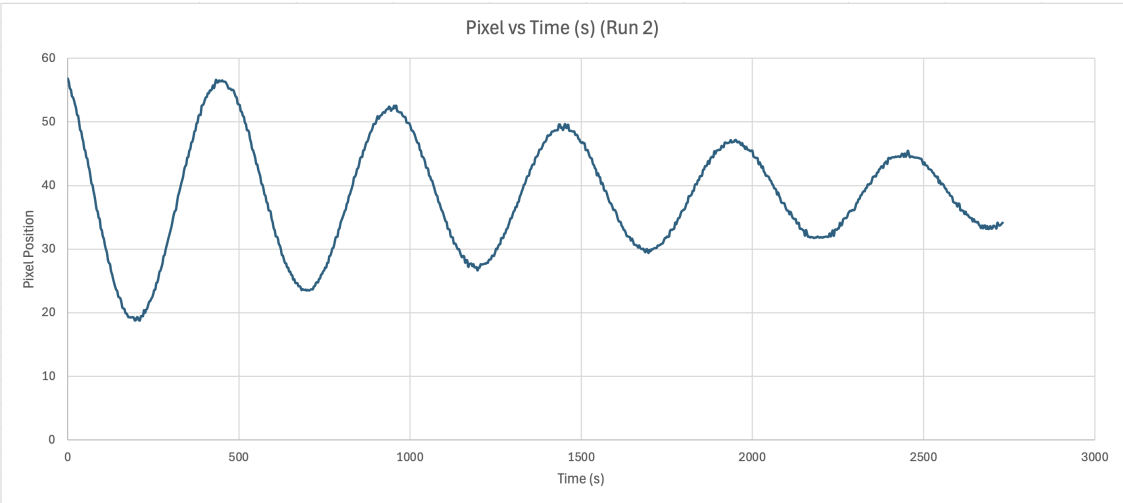


Figure 3: Pixel position of the reflected laser beam as a function of time for Run 2. Similar oscillatory behaviour was observed after repositioning the external masses.

3.2 Peak Measurements and Oscillation Period

Successive maxima of the oscillatory motion were identified in the Run 1 dataset in order to determine the oscillation period of the torsion system. The time and pixel position of each peak were recorded and the oscillation amplitude was determined relative to the equilibrium position.

Table 1: Measured peak positions of the torsional oscillations.

Peak	Time (s)	Pixel Position	Amplitude A
1	153.373	70.785	8.793
2	643.543	68.879	6.887
3	1145.740	67.634	5.642
4	1626.892	66.487	4.495
5	2129.159	65.615	3.623
6	2619.460	64.794	2.802

The oscillation period was obtained from the time difference between successive peaks.

Table 2: Oscillation periods calculated from successive peak intervals.

Peak Interval	Period T (s)
1–2	490.17
2–3	502.20
3–4	481.15
4–5	502.27
5–6	490.30

From these measurements the mean oscillation period of the torsion system was determined to be

$$T = (493.22 \pm 0.13)\,\mathrm{s}. \tag{11}$$

3.3 Damping Behaviour

The damping behaviour of the torsion balance was investigated by examining the decay of oscillation amplitude with time. The oscillation amplitude A decreases approximately exponentially with time and can therefore be described by

$$A(t) = A_0 e^{-\beta t}. \tag{12}$$

Taking the natural logarithm gives

$$\ln A = \ln A_0 - \beta t. \tag{13}$$

The natural logarithm of the measured amplitudes is listed in Table 3.

Table 3: Natural logarithm of oscillation amplitudes used for damping analysis.

Peak	Time (s)	$\ln(A)$
1	153.373	2.174
2	643.543	1.930
3	1145.740	1.730
4	1626.892	1.503
5	2129.159	1.287
6	2619.460	1.030

A plot of $\ln(A)$ as a function of time is shown in Fig. 4. The approximately linear behaviour confirms the exponential decay of the oscillation amplitude.

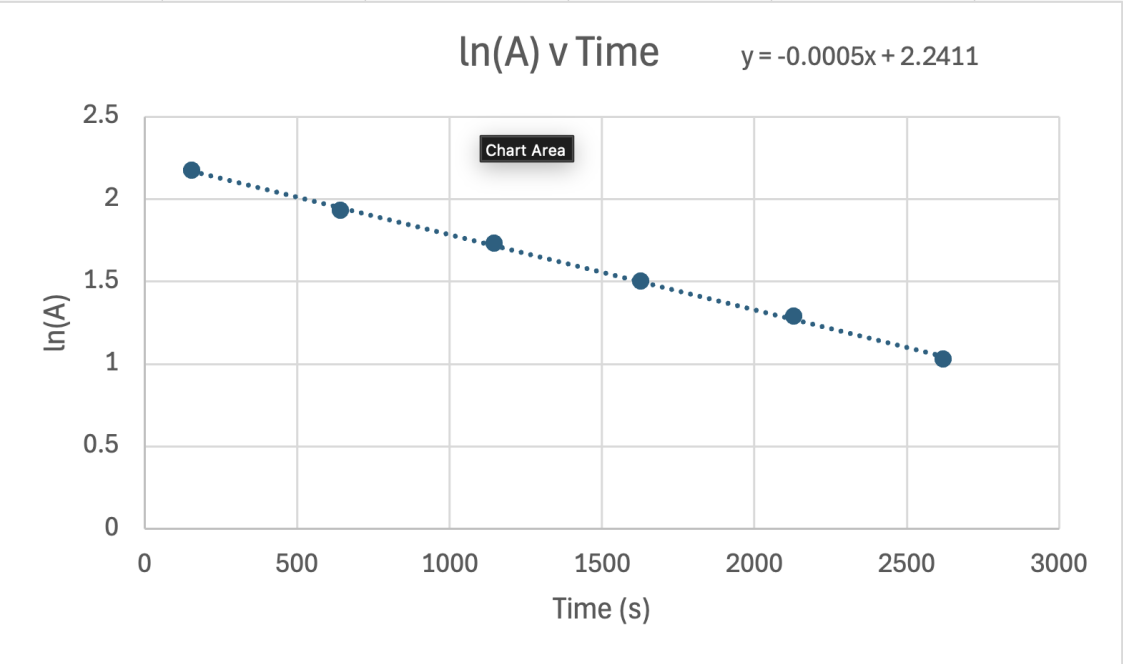


Figure 4: Natural logarithm of oscillation amplitude as a function of time. The linear trend indicates exponential damping of the torsional oscillations.

The best fit line to the data has the form

$$\ln(A) = -0.0005\,t + 2.2411, \tag{14}$$

giving a damping constant

$$\beta = 5.0 \times 10^{-4}\,\mathrm{s}^{-1}. \tag{15}$$

3.4 Measured Angular Deflection

The equilibrium displacement of the torsion balance was determined from the shift in the pixel position of the reflected laser beam. The measured change in pixel position was

$$\Delta p = 23.48 \text{ pixels}. \tag{16}$$

Using the detector pixel size of 8.0×10^{-4} m, the corresponding displacement on the detector was

$$x = 1.88 \times 10^{-2}\,\text{m}. \tag{17}$$

The distance between the mirror and the photodiode array was measured to be

$$D = (0.81 \pm 0.01)\,\text{m}. \tag{18}$$

Using the optical lever relation, the angular deflection of the torsion balance was determined to be

$$\Delta\theta = 1.16 \times 10^{-2}\,\text{rad}. \tag{19}$$

3.5 Summary of Measured Quantities

The experimentally measured parameters used in the analysis are summarized in Table 4.

Table 4: Measured experimental parameters used in the analysis.

Parameter	Value
Large mass M	1.5 kg
Small mass m	0.020 kg
Distance d or L	0.050 m
Pixel size	8.0×10^{-4} m
Laser distance D	(0.81 ± 0.01) m
Mean oscillation period T	(493.22 ± 0.13) s
Angular deflection $\Delta\theta$	1.16×10^{-2} rad

3.6 Moment of Inertia of the Torsion Balance

The torsion balance consisted of two identical small masses attached to the ends of a horizontal beam. Assuming the mass of the beam itself is negligible compared with the attached masses, the moment of inertia of the rotating system about the torsion axis is given by

$$I = 2mL^2, \tag{20}$$

where m is the mass of each small sphere and L is the distance from the torsion axis to the centre of each sphere. The derivation of this relation is provided in Appendix A.

Using the measured experimental values for m and L, the moment of inertia of the torsion system was determined to be

$$I = 1.0 \times 10^{-4}\ \mathrm{kg\,m^2}. \tag{21}$$

3.7 Torsion Constant of the Fibre

The torsion constant of the fibre cannot be measured directly and must instead be determined from the oscillation period of the torsion balance. For small angular oscillations, the system behaves as a torsional harmonic oscillator with period

$$T = 2\pi\sqrt{\frac{I}{k}}, \tag{22}$$

where I is the moment of inertia and k is the torsion constant of the fibre. Rearranging gives

$$k = \frac{4\pi^2 I}{T^2}. \tag{23}$$

The derivation of this relation is provided in Appendix B. Substituting the measured oscillation period $T = (493.22 \pm 0.13)$ s and the calculated moment of inertia yields

$$k = 1.62 \times 10^{-8}\ \mathrm{N\,m\,rad^{-1}}. \tag{24}$$

3.8 Separation Parameter

The gravitational interaction between each large mass and the corresponding small mass depends on the effective centre-to-centre separation between the masses. In this experiment the separation parameter b was determined from the radius of the large sphere R and one quarter of the box thickness t,

$$b = R + \frac{t}{4}. \tag{25}$$

Using $R = 0.03193$ m and $t = 0.0301$ m gives

$$b = 0.039455\ \mathrm{m}. \tag{26}$$

3.9 Angular Deflection from Optical Lever Geometry

The angular displacement of the torsion balance was determined from the displacement of the reflected laser beam on the photodiode array. The relationship between the angular deflection and the displacement on the detector follows from the optical lever geometry,

$$\Delta\theta = \frac{x}{2D}, \tag{27}$$

where x is the displacement of the laser spot on the detector and D is the distance between the mirror and the detector. The derivation of this relation is provided in Appendix C.

Using the measured displacement $x = 1.88 \times 10^{-2}$ m and detector distance $D = 0.81$ m gives

$$\Delta\theta = 1.16 \times 10^{-2} \text{ rad.} \tag{28}$$

3.10 Determination of the Gravitational Constant

The gravitational attraction between the large and small masses produces a torque on the torsion balance. At equilibrium this gravitational torque is balanced by the restoring torque of the torsion fibre,

$$k\theta = 2L\frac{GMm}{b^2}. \tag{29}$$

Solving for the gravitational constant gives

$$G = \frac{k\theta b^2}{2MmL}. \tag{30}$$

The derivation of this expression is provided in Appendix D. Substituting the experimentally measured parameters yields

$$G = 9.77 \times 10^{-11} \text{ m}^3\,\text{kg}^{-1}\,\text{s}^{-2}. \tag{31}$$

3.11 Uncertainty in the Gravitational Constant

The uncertainty in the calculated value of G was determined using standard uncertainty propagation. For a quantity of the form

$$G = \frac{k\theta b^2}{2MmL}, \tag{32}$$

the fractional uncertainty is

$$\left(\frac{\sigma_G}{G}\right)^2 = \left(\frac{\sigma_k}{k}\right)^2 + \left(\frac{\sigma_\theta}{\theta}\right)^2 + \left(\frac{2\sigma_b}{b}\right)^2 + \left(\frac{\sigma_M}{M}\right)^2 + \left(\frac{\sigma_m}{m}\right)^2 + \left(\frac{\sigma_L}{L}\right)^2. \tag{33}$$

The detailed derivation of this uncertainty propagation is provided in Appendix E. Using the measured uncertainties for each parameter gives

$$\sigma_G = 1.79 \times 10^{-12} \text{ m}^3\,\text{kg}^{-1}\,\text{s}^{-2}. \tag{34}$$

The final experimental result for the gravitational constant is therefore

$$G = (9.77 \pm 0.18) \times 10^{-11} \text{ m}^3\,\text{kg}^{-1}\,\text{s}^{-2}. \tag{35}$$

4 Discussion

4.1 Summary of Experimental Results

An experimental value of the gravitational constant was obtained as

$$G = (9.77 \pm 0.18) \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}},$$

which differs from the accepted value of $6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ by approximately 46.4%. This value was determined using the measured oscillation period of the torsion balance and the angular deflection produced by the nearby large masses.

The oscillation period of the torsion balance was measured to be $T = (493.22 \pm 0.13)$ s, while the angular deflection obtained from the optical lever measurement was $\Delta\theta = 1.16 \times 10^{-2}$ rad. These quantities were used to determine the torsion constant of the fibre and the gravitational torque acting on the torsion balance.

The recorded oscillations exhibited clear damping behaviour, with the oscillation amplitude decreasing approximately exponentially with time. The damping constant obtained from the logarithmic amplitude analysis was $\beta = 5.0 \times 10^{-4}\ \mathrm{s^{-1}}$, consistent with energy dissipation in the torsion fibre and air resistance acting on the suspended system.

4.2 Comparison with the Accepted Value of G

The currently accepted value of the gravitational constant is

$$G_{\text{accepted}} = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}. \tag{36}$$

Comparison with the experimentally determined value yields a percent difference of approximately 46%. Although this deviation is substantial, it is not unexpected for a laboratory implementation of the Cavendish experiment. The gravitational force between laboratory-scale masses is extremely small, typically on the order of 10^{-10} N, making the measurement highly sensitive to small uncertainties in alignment, geometry, and environmental disturbances.

Even modern high precision measurements of the gravitational constant continue to show discrepancies larger than their reported uncertainties. As a result, G remains one of the least precisely known fundamental constants in physics.

4.3 Sources of Experimental Uncertainty

Several sources of random uncertainty affected the measurements obtained in this experiment. One important contribution arises from the determination of the oscillation period. The period was calculated from the time difference between successive peaks in the oscillatory motion, and small uncertainties in identifying the precise peak locations contribute to the uncertainty in the measured period.

Additional uncertainty arises from the determination of the angular displacement of the torsion balance. The angular deflection was obtained from the displacement of the reflected laser beam on the photodiode array. Uncertainty in the measured pixel displacement and in

the distance between the mirror and detector directly affects the calculated angular deflection. Because the gravitational constant depends linearly on this angle, uncertainty in the optical lever measurement propagates directly into the final value of G.

Uncertainties in the measured masses and geometric parameters also contribute to the overall uncertainty. In particular, the effective separation distance between the interacting masses enters the gravitational force through an inverse square dependence, making the calculated value of G especially sensitive to small uncertainties in this distance.

4.4 Systematic Errors and Experimental Limitations

In addition to random uncertainties, several systematic effects likely contributed to the observed deviation from the accepted value of G. One significant source of systematic error arises from the determination of the separation distance between the large and small masses. Because the gravitational force varies as $1/r^2$, even small errors in the effective centre-to-centre separation can lead to relatively large deviations in the calculated value of the gravitational constant.

Misalignment of the masses may also introduce systematic error. If the large spheres are not positioned symmetrically relative to the small masses, the resulting gravitational torque will differ from the idealized geometry assumed in the analysis. This misalignment can alter the magnitude of the torque acting on the torsion balance and therefore affect the calculated value of G.

Environmental disturbances represent another important limitation of the experiment. The torsion balance apparatus is extremely sensitive to external vibrations and mechanical disturbances. The experiment was performed in a laboratory environment that was not vibration free, meaning that small vibrations from building motion, nearby activity, or air movement could influence the equilibrium position of the torsion balance. Because the gravitational forces being measured are extremely weak, even minor disturbances can produce measurable shifts in the laser position and therefore affect the calculated angular deflection.

Another limitation arises from the assumption that the torsion fibre behaves as a perfectly linear torsional spring. In practice, real fibres may exhibit small nonlinearities, internal friction, or temperature dependent behaviour, which can influence the oscillation dynamics of the system and slightly alter the calculated torsion constant.

4.5 Possible Improvements

Several improvements could increase the precision of the experiment. Allowing additional time for the torsion balance to reach equilibrium before recording measurements would reduce transient oscillations and improve the determination of the equilibrium angular displacement.

Improved measurement of the separation distance between the masses would also reduce systematic uncertainty. Using more precise alignment procedures or mechanical measurement tools could help ensure that the geometry of the apparatus is accurately known.

The experiment could also benefit from improved isolation of the apparatus from environmental disturbances. Performing the measurement in a vibration isolated environment or within a sealed enclosure would reduce the influence of air currents and mechanical vibrations.

Finally, automated data acquisition and peak detection methods could improve the accuracy of oscillation period measurements by reducing uncertainty associated with manual identification of oscillation peaks.

Overall, despite the relatively large deviation from the accepted value, the experiment successfully demonstrates the fundamental principle of the Cavendish torsion balance and highlights the experimental challenges associated with measuring extremely weak gravitational forces between the laboratory masses.

5 Conclusion

In this experiment, a Cavendish torsion balance was used to measure the gravitational attraction between laboratory scale masses and determine an experimental value of the gravitational constant. The oscillation period of the torsion system and the angular deflection produced by nearby large masses were measured using an optical lever system consisting of a laser and photodiode array. These measurements allowed the torsion constant of the fibre and the gravitational torque acting on the balance to be determined.

From these measurements, the gravitational constant was found to be

$$G = (9.77 \pm 0.18) \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}. \tag{37}$$

This result differs from the accepted value of $6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ by approximately 46%. The discrepancy is largely attributed to the extreme sensitivity of the torsion balance apparatus to small uncertainties in geometric parameters, alignment of the masses, and environmental disturbances. In particular, the apparatus was highly sensitive to mechanical vibrations within the laboratory environment, which was not completely vibration free. Because the gravitational forces involved are extremely small, even minor disturbances can influence the measured angular displacement and therefore affect the calculated value of G.

Despite these limitations, the experiment successfully demonstrates the fundamental principle of the Cavendish torsion balance and illustrates how gravitational interactions between small masses can be measured experimentally. The experiment also highlights the inherent difficulty of determining the gravitational constant, which remains one of the least precisely known fundamental constants in physics.

AI Acknowledgment

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A Appendix

A.1 Appendix A: Moment of Inertia of the Torsion Balance

The moment of inertia of a rigid body about a rotation axis is defined as

$$I = \sum_i m_i r_i^2, \tag{38}$$

where m_i is the mass of the ith object and r_i is its distance from the rotation axis.

In the Cavendish torsion balance, two identical small spheres of mass m are attached to the ends of a horizontal beam. Each sphere is located a distance L from the torsion fibre axis. Treating the spheres as point masses relative to the rotation axis, the total moment of inertia of the system is therefore

$$I = mL^2 + mL^2. \tag{39}$$

Combining the contributions from both masses gives

$$I = 2mL^2. \tag{40}$$

This expression was used in Eq. (20) of the analysis to determine the moment of inertia of the torsion system.

A.2 Appendix B: Derivation of the Torsional Oscillation Period

When the torsion balance is rotated by a small angular displacement θ, the torsion fibre produces a restoring torque proportional to the angular displacement,

$$\tau = -k\theta, \tag{41}$$

where k is the torsion constant of the fibre.

Applying Newton's second law for rotational motion,

$$\tau = I\frac{d^2\theta}{dt^2}, \tag{42}$$

gives

$$I\frac{d^2\theta}{dt^2} = -k\theta. \tag{43}$$

Rearranging yields the differential equation

$$\frac{d^2\theta}{dt^2} + \frac{k}{I}\theta = 0, \tag{44}$$

which is the equation of simple harmonic motion. The angular frequency of the oscillation is therefore

$$\omega = \sqrt{\frac{k}{I}}. \tag{45}$$

The oscillation period is related to the angular frequency by

$$T = \frac{2\pi}{\omega}. \tag{46}$$

Substituting for ω gives

$$T = 2\pi\sqrt{\frac{I}{k}}. \tag{47}$$

Rearranging this expression yields the relation used in the analysis,

$$k = \frac{4\pi^2 I}{T^2}. \tag{48}$$

A.3 Appendix C: Optical Lever Geometry

The angular deflection of the torsion balance was determined using an optical lever consisting of a laser beam reflected from a mirror mounted on the torsion balance.

If the mirror rotates by a small angle $\Delta\theta$, the reflected beam rotates by twice this angle due to the law of reflection. The total angular deflection of the beam is therefore

$$\Delta\phi = 2\Delta\theta. \tag{49}$$

If the detector is located a distance D from the mirror, the displacement of the laser spot on the detector is approximately

$$x = D\,\Delta\phi. \tag{50}$$

Substituting $\Delta\phi = 2\Delta\theta$ gives

$$x = 2D\,\Delta\theta. \tag{51}$$

Solving for the angular deflection yields

$$\Delta\theta = \frac{x}{2D}. \tag{52}$$

This relation was used in Eq. (27) of the analysis to determine the angular deflection of the torsion balance from the measured laser displacement.

A.4 Appendix D: Derivation of the Cavendish Expression for G

The gravitational force between a large mass M and a small mass m separated by a distance b is given by Newton's law of gravitation,

$$F = \frac{GMm}{b^2}. \tag{53}$$

This force acts at a distance L from the torsion fibre axis and therefore produces a torque

$$\tau = LF. \tag{54}$$

Substituting for F gives

$$\tau = L\frac{GMm}{b^2}. \tag{55}$$

Because two identical mass pairs act on opposite sides of the torsion balance, the total gravitational torque is

$$\tau_{\text{grav}} = 2L\frac{GMm}{b^2}. \tag{56}$$

At equilibrium this torque is balanced by the restoring torque of the torsion fibre,

$$\tau_{\text{torsion}} = k\theta. \tag{57}$$

Setting these equal gives

$$k\theta = 2L\frac{GMm}{b^2}. \tag{58}$$

Solving for the gravitational constant yields

$$G = \frac{k\theta b^2}{2MmL}. \tag{59}$$

This expression was used in Eq. (30) of the analysis to determine the experimental value of the gravitational constant.

A.5 Appendix E: Uncertainty Propagation for G

The gravitational constant was calculated using

$$G = \frac{k\theta b^2}{2MmL}. \tag{60}$$

Taking the natural logarithm of both sides gives

$$\ln G = \ln k + \ln \theta + 2\ln b - \ln M - \ln m - \ln L. \tag{61}$$

Differentiating yields

$$\frac{dG}{G} = \frac{dk}{k} + \frac{d\theta}{\theta} + 2\frac{db}{b} - \frac{dM}{M} - \frac{dm}{m} - \frac{dL}{L}. \tag{62}$$

Assuming independent uncertainties and adding contributions in quadrature gives

$$\left(\frac{\sigma_G}{G}\right)^2 = \left(\frac{\sigma_k}{k}\right)^2 + \left(\frac{\sigma_\theta}{\theta}\right)^2 + \left(\frac{2\sigma_b}{b}\right)^2 + \left(\frac{\sigma_M}{M}\right)^2 + \left(\frac{\sigma_m}{m}\right)^2 + \left(\frac{\sigma_L}{L}\right)^2. \tag{63}$$

This relation was used in the analysis to determine the uncertainty in the experimentally measured value of the gravitational constant.